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Dimension of Greater India in the early Mesozoic: Paleomagnetic constraints from Triassic sediments in the Tethyan Himalaya

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ABSTRACT

The northern dimension of Greater India, defined as the continental part of India from which the Himalayan fold–thrust belt was derived, is difficult to determine due to major tectonic deformations of the Himalaya during the Tertiary caused by the India–Eurasian collision. Paleomagnetic data from the Tethyan Himalaya can potentially help determine this dimension. However, the tectonic deformations also led to widespread remagnetization of rocks there and thus previously published paleomagnetic poles from the lower Mesozoic Himalayan rocks may not accurately reflect their original paleomagnetic directions. To better understand the Triassic history of India in a Gondwana configuration, we report new paleomagnetic data (after tilt correction $D = 345.5°$, $I = -43.0°$, $\alpha 95 = 12.0°$, $k = 17.3$, $N = 10$ sites and the corresponding paleopole at 34.5°N, 282.2°E with $A_{95} = 11.7°$) of Triassic limestone and sandstone from the Tulong Group in the Tethyan Himalaya. The primary features of the magnetization are supported by the positive fold test. Our new Triassic paleomagnetic pole overlaps with 230–250 Ma poles from India, suggesting the close affinity between the Tethyan Himalaya and cratonic India since the Triassic. Assuming that the Tethyan Himalaya was a part of the Great India since Triassic, the differences between the Tethyan Himalaya paleolatitude obtained by this study and those previously reported for the cratonic India imply ~1300 km dimension of the Tethyan Himalaya has been absorbed after the Eurasia–India collision.

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1. Introduction

The Tethyan Himalaya is a fold–thrust belt that was derived from the northernmost continental margin of the Indian continent, and is separated from the Lesser Himalaya and the Indian continent by a crystalline belt of the Greater Himalaya (GH). The GH in turn is bounded by the synchronous Miocene top-to-the-north Southern Tibet Detachment System (STDS) at the top-to-the-south Main Central Thrust (MCT) at the base (e.g., Yin, 2006; Aitchison et al., 2007). Although debate on the amount of displacement and timing of activity of these structures makes it difficult to determine the paleogeographic affinity and dimension of the Tethyan Himalaya with respect to the cratonic India (e.g., Burchfiel et al., 1992; DeCelles et al., 2000; Myrow et al., 2003; Torsvik et al., 2009) (Fig. 1A), the Tethyan Himalaya is still the ideal object to constrain the initiation age of Eurasia–India collision (e.g., Yin and Harrison, 2000; Wang et al., 2002, 2011; Aitchison et al., 2007). However, processes of the initial Eurasia–India collision remain controversial because ranges of the initiation age are from the Late Cretaceous (e.g., Willems et al., 1996; Yin and Harrison, 2000; Cai et al., 2011) to the Eocene (e.g., Zhu et al., 2005a; Wang et al., 2011).

Among previous investigations, the sedimentary strata of the Late Cretaceous and Eocene in the Tethyan Himalaya have been intensively studied geologically (e.g., Aitchison et al., 2007; Najman et al., 2010). However, the early Mesozoic of the Tethyan Himalaya remains poorly understood. Whether there is a close affinity between the Tethyan Himalaya and the cratonic India in the Mesozoic is a key to define the initial Eurasia–India collision (Patzelt et al., 1996; Chen et al., 2010; Sun et al., 2010; Najman et al., 2010). Several researchers have proposed that the Tethyan Himalaya should be located in the northern part of Greater India since the Paleozoic (Cocks and Fortey, 1988; DeCelles et al., 2000; Yoshida and Upreti, 2006). Recent workers have challenged

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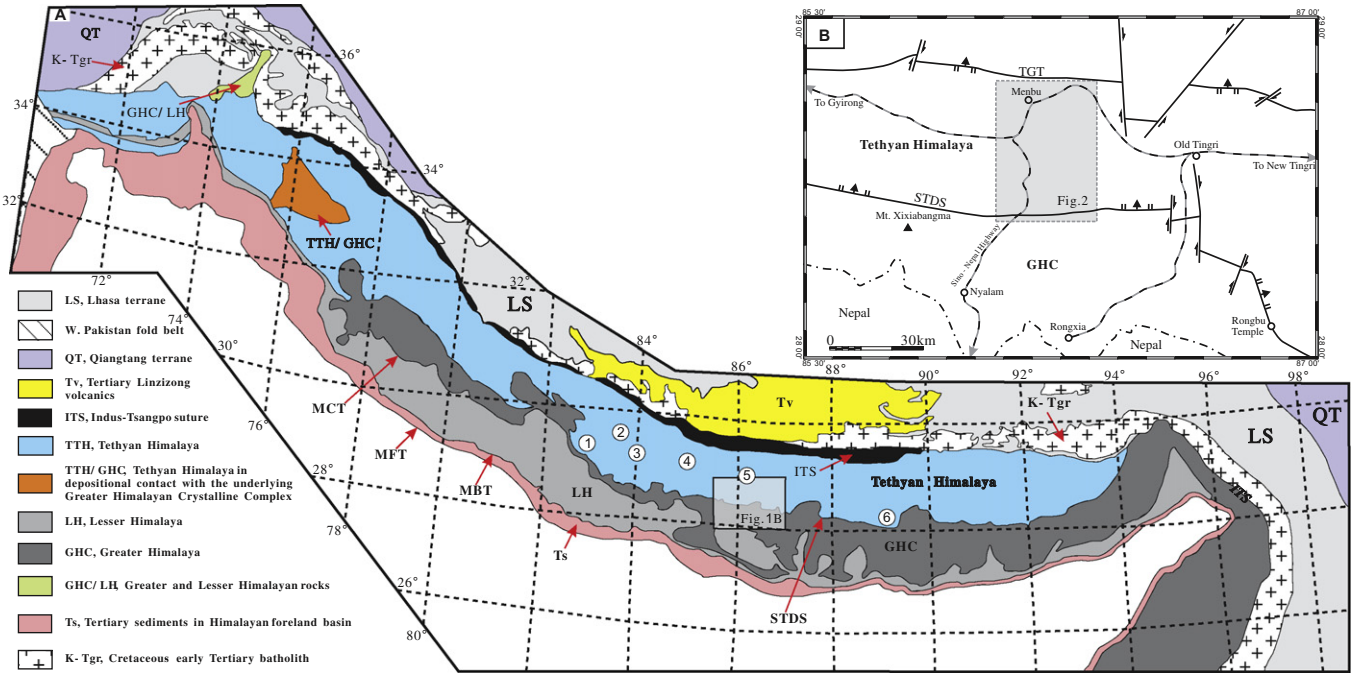


Fig. 1. Simplified geological map of the Himalayan belt (A was modified from Crouzet et al., 2003 and Yin, 2006; B was modified from Zhu et al., 2002; Wang et al., 2005 and Hu et al., 2008). ① – Western Dolpo (Crouzet et al., 2003), ② – Thakkhola (Klootwijk and Bingham, 1980), ③ – Manang (Appel et al., 1991), ④ – Shiar (Schill et al., 2002), ⑤ – Tingri (Besse et al., 1984; Tong et al., 2008), ⑥ – Gamba (Yi et al., 2011). TGT, Tingri–Gamba Thrust; MCT, Main Central Thrust; MBT, Main Boundary Thrust; MFT, Main Frontal Thrust; STDS, Southern Tibet Detachment System; ITS, Indus–Tsangpo suture. The main B's legends are in Fig. 2.

this continuous margin model with new geochronology, isotopic and petrology data that suggest that the Tethyan Himalaya in the Mesozoic might be preserved as an independent terrane located at the south of the Lhasa terrane prior to the Eurasia–India collision rather than northern extension of the Indian continent (e.g., Li et al., 2010; Liu et al., 2010).

Paleomagnetism from the Himalayan fold-and-thrust belts can potentially provide an independent reference frame that helps understand the affinity between the Tethyan Himalaya and cratonic India and quantify the dimension of the Greater India prior to the collisional deformation (e.g., Torsvik et al., 2009). However, primary remanence preserved in the Mesozoic rocks is very sparse although sometimes with a positive fold test result because of serious natural conditions and pervasive remagnetization event reported from the Mesozoic rocks in the Tethyan Himalaya, so that the bulk of the former Triassic poles in the Tethyan Himalaya plot at more southerly latitudes than that of the cratonic India, which is at odd with the Late Cretaceous polar wander pattern (Klootwijk and Bingham, 1980; Besse et al., 1984; Appel et al., 1991; Patzelt et al., 1996; Schill et al., 2002; Crouzet et al., 2003; Tong et al., 2008; Torsvik et al., 2008). This study focuses on paleomagnetic analyses of the Lower–Middle Triassic rocks that were collected from the Tingri to Nyalam areas of the Tethyan Himalaya (Fig. 1). Our results show the Tethyan Himalaya was located at the north part of the Indian continent since the Mesozoic with a dimension of ~1300 km prior to the Eurasia–India collision.

2. Geological setting and samples

The Tethyan Himalaya is located at south of the Indus–Tsangpo suture zone (ITS) and north of the STDS, with abundant Late Paleozoic and Mesozoic strata in this region (Fig. 1A; e.g., BGMRX, 1993; Garzanti et al., 1999; Zhu et al., 2002; Hu et al., 2008). These strata continue up to Eocene (Table 1; Wan et al., 2000; Hu et al., 2008), except for the unconformity between the Lower Carboniferous and Lower Permian (BGMRX, 1993; Li et al., 1997; Garzanti, 1999; Zhu et al., 2002). During the Middle Cretaceous, the northward-directed subduction of the Neo-Tethyan oceanic crust beneath the Lhasa terrane in the southern margin of the Eurasia Plate led to the closure of the Neotethyan ocean and the formation of the ITS (e.g., Allégre et al., 1984; Dewey et al., 1988; Yin and Harrison, 2000; Najman et al., 2010). Furthermore, continent–continent collision resulted in the onset of Himalayan orogenesis, including mainly the STDS, MCT, Main Boundary Thrust (MBT), Main Frontal Thrust (MFT), and Greater Counter Thrust (Fig. 1A; e.g., Yin and Harrison, 2000; Yin, 2006; Hu et al., 2008). Our study area lies above the STDS and in an intra-TH thrust, the north-dipping Tingri–Gamba Thrust (TGT) (Fig. 1B). The Tingri–Gamba Thrust is believed to have formed during the Cenozoic (Zhu et al., 2002; Yin, 2006; Hu et al., 2008).

Previous mapping has determined the ages of the stratigraphy of the study area (Wan et al., 2000; Sakagami et al., 2006; Hu et al., 2008; Brühwiler et al., 2010; Forel et al., 2011). Table 1 lists the most recent stratigraphic scheme: (1) the Middle–Upper Permian Selong Group, with basalts alternating with shales and sandstones; (2) the Lower to Upper Triassic Tulong group, with gray laminated limestone alternating with shales and sandstones; (3) the Upper Triassic Quronggongba Formation, with the shales interbedded with sandstones; (4) the Upper Triassic Derirong Formation, with quartz sandstones; (5) the Lower Jurassic Pupuga Formation, with bioclastic limestones; (6) the Middle to Upper Jurassic Nieniexiongla and Lanongla Formation, with bioclastic limestones and shales; (7) the Upper Jurassic Menkadun Formation, with quartzy sandstones; and (8) the Upper Jurassic–Lower Cretaceous Gucuo Formation, with the shales interbedded with sandstones. These rocks were deposited continuously, without hiatus or unconformity (Zhu et al., 2002; Yin et al., 2007; Hu et al., 2008). Major folding and regional deformation occurred probably between Late Cretaceous and Tertiary (BGMRX, 1993).

We collected a total of 129 individual oriented core samples from 11 Triassic sites with a portable petro-powered drill along

Table 1
Previous proposal for the lithostratigraphic subdivision and dating of the Lower Carboniferous – Cretaceous strata in the Tethyan Himalaya.

Age		BGMRX, 1993	This study	Abbreviate
Cretaceous		Gamba Gp.	Gamba Gp.	K_{2g}
			Gucuo Fm.	K_{1gc}
Jurassic	J_3	Menkadun Fm.	Menkadun Fm.	J_{3m}
	J_2	Nieniexiongla Fm.	Lanongla Fm.	J_{2l}
			Nieniexiongla Fm.	J_{2n}
	J_1		Pupuga Fm.	J_{1p}
Triassic	T_3	Derirong Fm.	Derirong Fm.	T_{3d}
		Quronggongba Fm.	Quronggongba Fm.	T_{3q}
	T_2	Tulong Gp.	Tulong Gp*	$T_{(1-3)t}$
	T_1			
Permian	P3		Selong Gp.	$P_{(2-3)s}$
	P2	Selong Gp.		
	P1	Jilong Fm.	Jilong Fm.	P_{1j}

* Gray and represent strata of sampled; Fm., formation; Gp., group. This scheme is from BGMRX (1993), Hu et al. (2008), Zhu et al. (2002) and Wang et al. (2005).

the Tingri–Nyalam part of the Sino-Nepal highway (Fig. 2), the Tulong Group sandstones and limestones (Figs. 3A and 3B). Some sites were oriented by both sun and magnetic compasses to evaluate any differences among the local declinations and the present International Geomagnetic Reference Field in the sampling area.

3. Rock magnetic study

3.1. Techniques

The acquisition of isothermal remanent magnetization (IRM) and the progressive thermal demagnetization of composite IRMs were used to identify the coercivity and unblocking temperature spectra of 2–3 representative samples from the Triassic rocks. A direct-current magnetic field was applied to acquire the IRM with a 2G Enterprises Pulse Magnetizer with a maximum field of up to 2700 mT. These specimens were also subjected to progressive thermal demagnetization at temperatures from room temperature to 690 °C at intervals of 50, 30 or 20 °C after applying a composite IRM, as described by Lowrie (1990). DC magnetic fields of 2700; 400; and 140 mT were applied to the specimen's *Z*-, *Y*- and *X*-axes, respectively. The remanent magnetizations were measured with a 2G Enterprises cryogenic magnetometer (2G-755) at the South China Sea Institute of Oceanology, Chinese Academy of Sciences.

3.2. Results

The results of these experiments are shown in the IRM acquisition curves and the demagnetization curves of the IRM (Fig. 4). The IRM acquisition experiments and the thermal demagnetization of the composite IRMs showed that the sandstone and limestone samples are dominated by magnetite. Indeed, the IRM of the specimen usually reaches quasi-saturation below 400 mT, and the progressive thermal demagnetization of the soft, medium, and hard components shows only limited unblocking temperature range of about 500–580 °C (Fig. 4).

4. Paleomagnetic study

4.1. Techniques

In the laboratory, minicores (2.5 cm in diameter) were cut into 2.2-cm-long specimens. The magnetic measurements and demagnetization procedures were performed in a magnetically shielded room in the paleomagnetic laboratory of the South China Sea Institute of Oceanology (SCSIO), Chinese Academy of Sciences. All of the specimens were progressively thermally demagnetized with a thermal demagnetizer. Most of the specimens were thermally demagnetized over 12 temperature steps up to 500–695 °C within a nearly-zero-field laboratory-built furnace at SCSIO. The demagnetization intervals were as large as 50 or 100 °C at the lower temperatures and as small as 20 or 30 °C at the higher temperatures. All of the remanence measurements were performed on a 2G-755 cryogenic magnetometer at SCSIO. Both the demagnetizer and magnetometer were installed in a magnetically shielded space with the inside field minimized at <300 nT. During thermal demagnetization, the specimen orientation was successively inverted about the *Z*-axis to detect any systematic magnetization that could have resulted from the small, ~10 nT residual magnetic field in the Schonstedt furnace.

The demagnetization results were plotted as orthogonal vector diagrams (Zijderveld, 1967) and as equal-area projections. Both paleomagnetic directions were determined using principal component analysis (Kirshvink, 1980). The site-mean directions were calculated using Fisher (1953) statistics for the directional data. All of the interpretations and data processing were carried out using Paleomagnetism Data Analysis Version 4.2 software (available from Randy Enkin's website: http://gsc.nrcan.gc.ca/sw/paleo_e.php).

4.2. Demagnetization results

The demagnetization results for 52 of the 129 Triassic samples yield relatively simple demagnetization behavior after the removal of a soft component during the initial demagnetization (temperature steps of 100–150 °C, see Fig. 5). The directions of this soft component generally cluster near the geocentric axial field direction at the sampling area. Progressive thermal demagnetization to 600 °C, however, reveals two other components of magnetization in about half of the samples.An intermediate unblocking temperature component (A) was isolated in near all samples by best-fitting lines to demagnetization data between 300 and 450 °C. A high unblocking temperature component (B) was isolated between 500 and 620 °C.

The major rocks in the Tulong Group are gray limestone and sandstone (HT-sites and T1W). The direction of component A has approximately a NNE declination and a medial upward inclination *in situ* (Fig. 5 and Table 2). For component B, there are two groups: one was usually unblocked at 530–580 °C and had a NNE declination and a shallow downward inclination *in situ*; the other was usually unblocked at 520–620 °C and had a NNW declination and a shallow upward inclination *in situ*, especially in the SWW

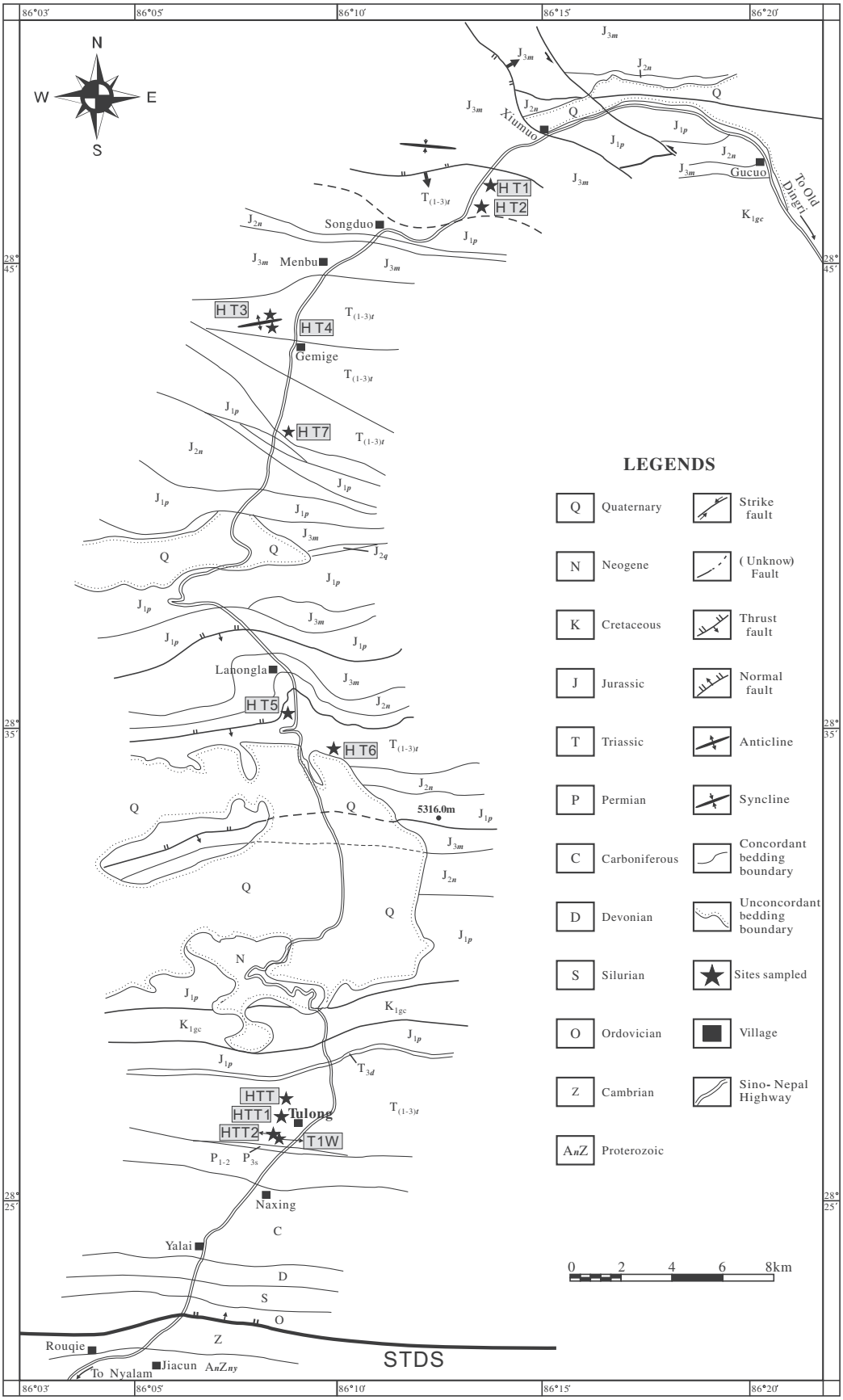


Fig. 2. Geological map and sampling sites of the study area in Nyalam (modified from Wang et al., 2005; Zhu et al., 2002).



Fig. 3. Photographs showing the Triassic strata sampled in the study area. (A) The gray sandstone of the Tulong Group in the Tulong country in the site HTT; (B) the gray limestone of the Tulong Group in the site HT7. The length of hammer is 18 cm.

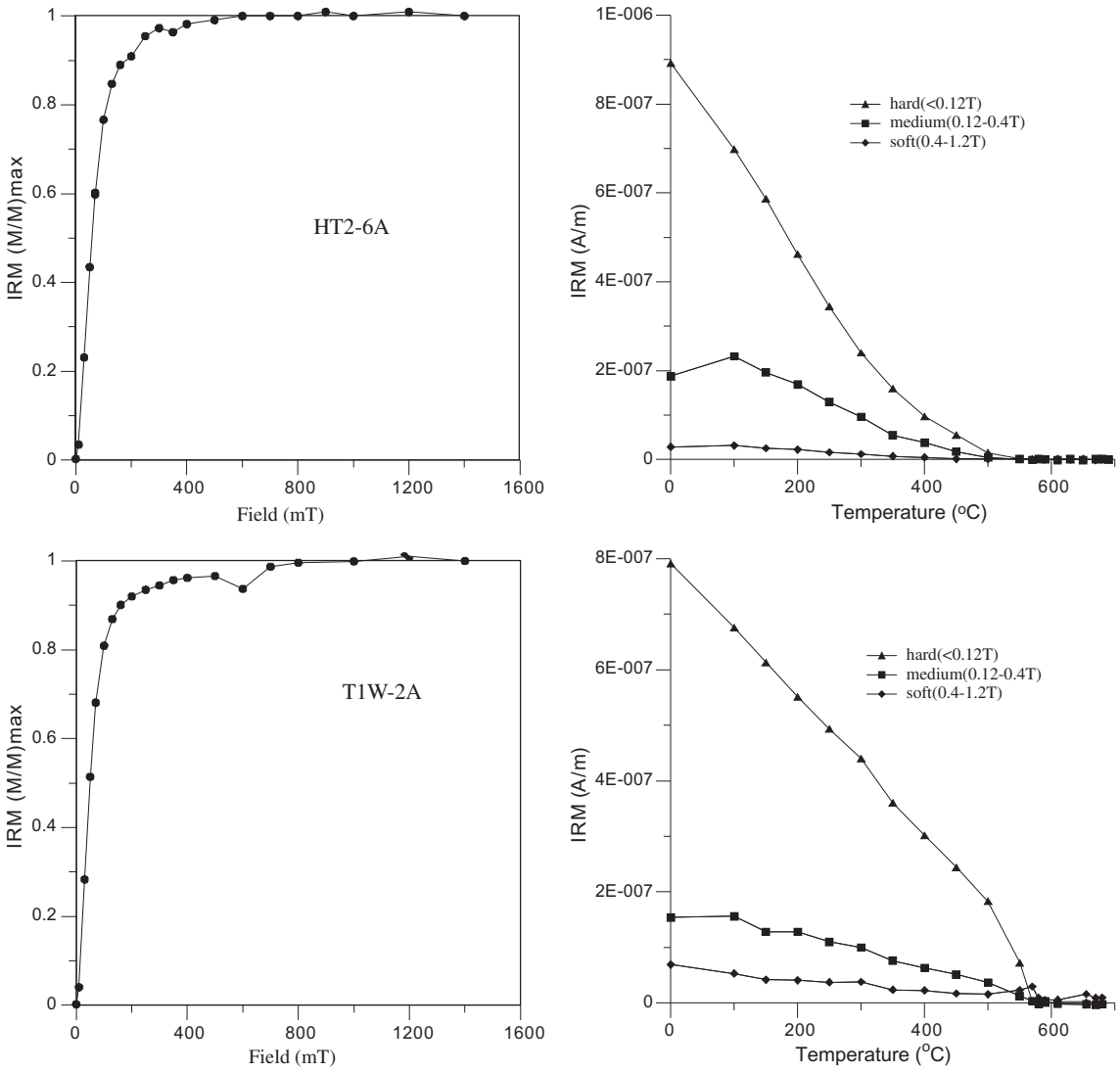


Fig. 4. Rock magnetic behavior of specimen HT2-6A and T1W-2A: (left) acquisition of isothermal remanent magnetization (IRM); (right) progressive thermal demagnetization curves of composite IRMs following Lowrie (1990).

declination at HT7 (Fig. 5, specimen HT7-28A). For site HT6, component B is difficult to interpret because of the highly scatter directions. Hence this site was disregarded (Table 3).

Interestingly, the thermal demagnetization of the gray limestone samples is characterized by the appearance of a high-temperature (B) component only (Fig. 5, specimens T1W-1A and

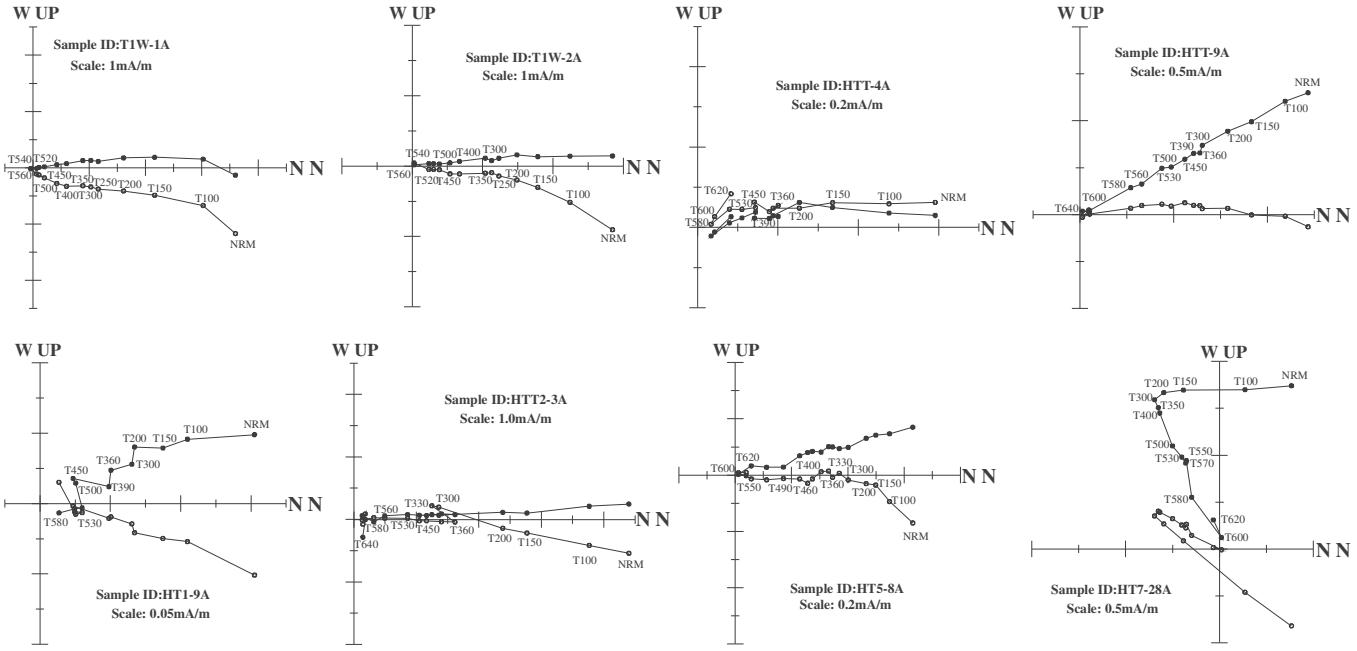


Fig. 5. Representative vector endpoint diagrams showing results of thermal demagnetization for samples in this study area. Directions are plotted in geographic coordinates. Solid circles and open circles represent the projection of the magnetization vector endpoint on the horizontal and vertical planes, respectively. NRM: Natural Remanent Magnetization. Temperatures shown are in degrees Celsius.

Table 2
Summary of the Triassic results of the A component from the Nyalam area in the Tethyan Himalaya.

Site ID	GPS coordinates		Bedding attitude		Mean direction							
	Latitude (°N)	Longitude (°E)	Strike (°)	Dip (°)	N/No	Geographic coordinates				Stratigraphic coordinates			
						D geo (°)	*I geo* (°)	*k*	*α95* (°)	*D str* (°)	*I str* (°)	*k*	*α95* (°)
HT1	28°45′45.2″	86°10′44.6″	307	40	0/7	–	–	–	–	–	–	–	–
HT2	28°45′45.8″	86°10′44.7″	307	40	5/6	358.2	44.4	19.0	18.0	13.7	19.2	19.0	18.0
HT3	28°42′50.4″	86°08′21.5″	139	61	8/11	355.2	48.6	16.6	14.0	277.8	44.9	16.6	14.0
HT4	28°42′54.7″	86°08′23.5″	276	35	4/9	11.4	58.5	14.8	24.7	9.1	23.1	14.8	24.7
HT5	28°35′03.9″	86°08′42.5″	274	36	8/9	359.9	39.6	17.3	13.7	0.9	3.7	17.3	13.7
HT6	28°34′25.2″	86°09′06.0″	274	49	6/9	341.9	28.1	28.3	12.8	343.5	−17.9	28.3	12.8
HT7	28°42′10.9″	86°08′38.0″	110	88	20/35	0.5	42.3	18.5	7.8	226.5	45.3	18.5	7.8
HTT	28°26′46.1″	86°09′02.4″	286	57	0/10	–	–	–	–	–	–	–	–
HTT1	28°26′42.0″	86°08′58.7″	286	38	7/13	1.2	−1.2	64.0	7.6	357.1	−37.6	64.0	7.6
HTT2	28°26′12.2″	86°08′16.1″	282	46	4/10	354.2	4.6	24.8	18.8	349.0	−38.8	24.8	18.8
T1W	28°26′12.2″	86°08′16.1″	284	69	6/10	0.8	0.0	20.9	15.0	340.8	−65.4	20.9	15.0
Mean component A from Triassic rocks					9/11	355.0	28.8	12.2	14.4	342.7	−7.5	2.4	39.6

Abbreviations are: site ID, site identification; strike/dip, strike azimuth and dip of bed; N/No, number of samples (sites) joined the calculation/demagnetized; *D geo*, *I geo* (*D str*, *I str*), declination and inclination of direction *in situ* (after tilt correction); *k*: precision parameter, *a95*: 95% confidence angle.

T1W-2A). Component B in the limestone samples was usually completely unblocked at 560 °C. This component had a northward declination and a shallow downward inclination *in situ*.

5. Field tests for stability

Significant variation in the bedding attitudes (strikes from 110° to 307°) exists within the sampling area, allowing a meaningful fold test to be conducted. Components A and B could be isolated from the 11 Triassic sites (Fig. 5), and the mean directions are clustered. Component A has an *in situ* site mean direction of *D* (declination) = 355.0°, *I* (inclination) = 28.8°, *k* (precision parameter) = 12.2, and *α95* (95% confidence circle about the site mean direction) = 14.4° and a tilt-corrected mean direction of *D* = 342.7°, *I* = −7.5°, *k* = 2.4, and *α95* = 39.6° for the ten sites (with site HTT was disregarded) (Fig. 6A and Table 2). The obvious negative fold test results indicate a post-folding origin for the component A. However, some of the site-mean directions (sites HT5, HT6, HTT1, HTT2 and T1W) are significantly away from the present-day geomagnetic field (PGF) direction with shallower *in situ* inclinations, suggesting that component A was not entirely acquired during the recent time.

We interpret the component B as the characteristic remanent magnetization (ChRM). Ten out of the 11 Triassic sites provided a site mean direction of the ChRM with *D* = 347.1°, *I* = −9.7°, *k* = 4.4, and *α95* = 26.0° before and *D* = 345.5°, *I* = −43.0°, *k* = 17.3, and *α95* = 12.0° after unfolding (Fig. 6B and Table 3). Based on the fold test results, it was found that the optimum concentration of the precision parameter (*k*) was first achieved at 103.9 ± 8.3% unfolding in Watson and Enkin (1993) Test and the precision parameter ratio $ks/kg = 3.932 > F[18,18] = 2.217$ in the grouping of the ChRM at 95% confidence limits in McElhiny's fold Test (1964). Appling the test of McFadden (1990), the fold test results again are positive at more than 95% confidence level for the

Table 3
Summary of the Triassic results of the B component from the Nyalam area in the Tethyan Himalaya.

Site ID	Mean direction								
	N/No	Geographic coordinates				Stratigraphic coordinates			
		D geo (°)	*I geo* (°)	*k*	*α95* (°)	*D str* (°)	*I str* (°)	*k*	*α95* (°)
HT1	4/7	15.6	4.1	37.8	15.1	11.2	−32.7	37.7	15.2
HT2	4/6	355.7	−27.5	55.8	12.4	327.0	−51.2	55.9	12.4
HT3	4/11	318.8	−30.8	16.4	23.4	346.5	−14.6	16.5	23.3
HT4	3/9	21.4	7.9	16.5	31.3	23.1	−26.3	16.5	31.4
HT5	8/9	351.6	−1.5	16.0	14.3	348.4	−36.5	16.0	14.3
HT6	0/9	–	–	–	–	–	–	–	–
HT7	8/35	239.1	−22.7	28.3	10.6	330.9	−52.1	28.3	10.6
HTT	6/10	346.5	−11.1	50.3	9.5	318.1	−54.8	50.4	9.5
HTT1	4/13	350.6	2.0	16.9	23.0	345.6	−31.8	16.9	23.0
HTT2	4/10	354.1	−4.4	147.1	7.6	345.2	−47.3	147.1	7.6
T1W	7/10	355.8	18.6	53.8	8.3	348.4	−46.6	53.8	8.3
Triassic Mean	10/11 (sites)	347.1	−9.7	4.4	26.0	345.5	−43.0	17.3	12.0

The abbreviations are the same as for Table 2. Fold test of all Triassic sites except for HT6:
(1) Watson and Enkin's Test (1993): the optimum concentration is achieved at 103.9 ± 8.3% unfolding;
(2) McElhiny's Fold Test (1964): $N = 10$, $ks/kg = 3.932 > F[18,18] = 2.217$ of the grouping of the ChRM, at 95% confidence limits, indicative of a positive fold test.
(3) McFadden's Fold Test (1990) using definition 2 of SCOS: $N = 10$, *in situ* $\xi_2 = 6.523$, Tilt-corrected $\xi_2 = 1.545$, statistical threshold $\xi = 3.685$ at 95% confidence level. The SCOS values after tilt-correction are far less than the critical values at 95% confidence level, so there is no reason to reject the hypothesis that the ChRM was acquired before the folding.

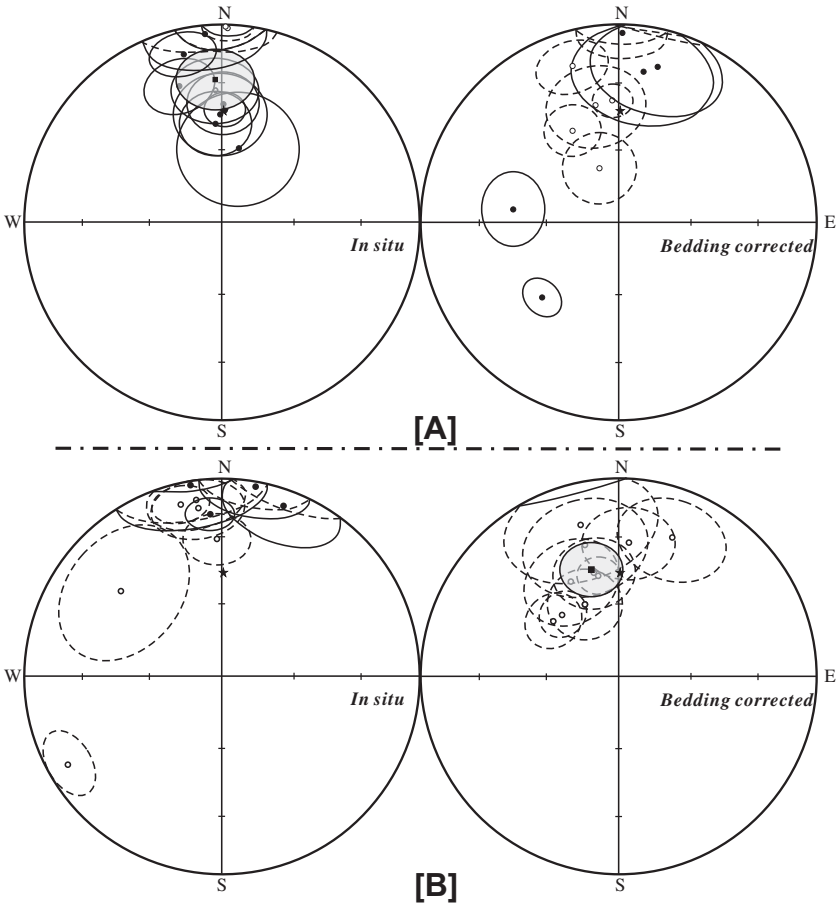


Fig. 6. Stereographic projection (upper hemisphere) of the low temperature (A) and high temperature (B) components site mean directions. The overall mean direction is indicated. Star is projected present magnetic field. Open (solid) squares are projected in the upper (lower) hemisphere.

Triassic sections. In the *in situ* coordinates, the SCOS values ($\xi_2 = 6.523$) exceed the critical values ($\xi = 3.685$) at the 95% confidence level, so there is a clear correlation between the ChRM directions and tectonics, thus, it is unlikely that the ChRM was acquired with bedding attitudes. Conversely, in the unfolded coordinates, the SCOS values ($\xi_2 = 1.545$) are far less than the critical values at 95% confidence level, so there is no reason to reject the hypothesis that the ChRM was acquired before the folding.

6. Discussion

6.1. Age of magnetization components

Determining the age of magnetization is absolutely critical but not always straightforward. This has been the single most challenging problem in paleomagnetic research in Asia. Both overprinted and primary components are revealed in the results from this study. From 68 out of the 129 Triassic samples, the mean direction for component A (Table 2) clearly failed the fold test: $D = 358.6°$, $I = 40.6°$, $k = 9.0$, and $\alpha 95 = 6.2°$ before and $D = 328.3°$, $I = 35.2°$, $k = 1.9$, and $\alpha 95 = 17.9°$ after unfolding, showing a post-folding age. Thus, component A is a secondary remanence imposed after folding, probably a recent field overprint.

The B component is characterized by linear demagnetization vector segments and was usually isolated only after removal of the A component. As documented above, fold tests on the mean direction of component B is entirely positive. Thus, we have no reason not to believe that the characteristic direction associated with the B component is primary. The successful fold test and the fact that the ChRM is carried by magnetic mineral (magnetite) that is capable of preserving an early remanent magnetization over a long period of time, strongly suggest that the magnetization directions derived from the Triassic rocks are primary and suitable for calculating the corresponding paleomagnetic pole for the Tethyan Himalaya.

6.2. Tectonic implication: dimension of Greater India

As mentioned in the introduction, the Tethyan Himalaya plays a key role in reconstructing the original extent of Greater India (e.g., Klootwijk and Bingham, 1980; Ali and Aitchison, 2005; Aitchison et al., 2007). Several researchers have proposed that the Tethyan Himalaya should be located in the northern part of Greater India since the Paleozoic (Cocks and Fortey, 1988; DeCelles et al., 2000; Yoshida and Upreti, 2006; Yi et al., 2011). On the other hand, recent workers have suggested that the Tethyan Himalaya might be an independent terrane located at the south of the Lhasa terrane prior to the Eurasia–India collision during the Mesozoic (e.g., Li et al., 2010; Liu et al., 2010).

Four paleomagnetic studies of Triassic sedimentary rocks are available in the Tethyan Himalaya (Fig. 7 and Table 4). Klootwijk and Bingham (1980) proposed a paleolatitude of 27.4° ± 5.3°S for the Thakkhola region in the Late Triassic. Appel et al. (1991) reported a paleomagnetic result ($P\lambda = 34.6° \pm 4.9°S$) from the Upper Triassic Mukut limestone in the Manang area with positive fold test. Schill et al. (2002) determined a 45.3° ± 15.3°S paleolatitude for the Tethyan Himalaya in the Early-Middle Triassic without field tests to assess the reliability because of similar bedding at sampling sites of the Shiar Khola area (Table 4). Crouzet et al. (2003) obtained Lower–Middle Triassic paleomagnetic dataset from the western Dolpo area with the higher paleolatitude of 55.4° ± 17.1°S. These paleomagnetic data from the Tethyan Himalaya yield systematically more southerly latitudes than those of the cratonic India (Fig. 7; Torsvik et al., 2008). On the other hand, the Late Cretaceous and Paleocene paleolatitudes determined by Patzelt et al. (1996) and Yi et al. (2011), which are the best quality ($Q = 7$) (van der Voo, 1990) among the studies had been reported in the Late Cretaceous and Paleocene (Table 4; Besse et al., 1984; Tong et al., 2008; Najman et al., 2010), are more northerly than the cratonic India (Fig. 7). Because no suture has been identified to suggest Tethyan Himalaya was a separated terrane from Indian continent, the paleomagnetic results of Patzelt et al. (1996) and Yi et al. (2011) have suggested that Tethyan Himalaya was undoubtedly the northern extension of the Great India and that >1000 km of post-collisional crustal shortening occurred within the Himalayas since the Late Cretaceous.

Our Tulong Group pole with $Q = 6$ (without # 6 the reversals presence) yields lower latitudes (25.0° ± 11.7°S in Table 4) than expected mean paleolatitude (36.0° ± 3.6°S, 240 ± 10 Ma) for the Gondwana in cratonic India from 230 to 250 Ma (Torsvik et al., 2008). The Tethyan Himalaya pole implies 11° latitudes (~1300 km) more northerly than the Indian continent, which is consistent with the hypothesis that the Tethyan Himalaya should be the northern part of Great India

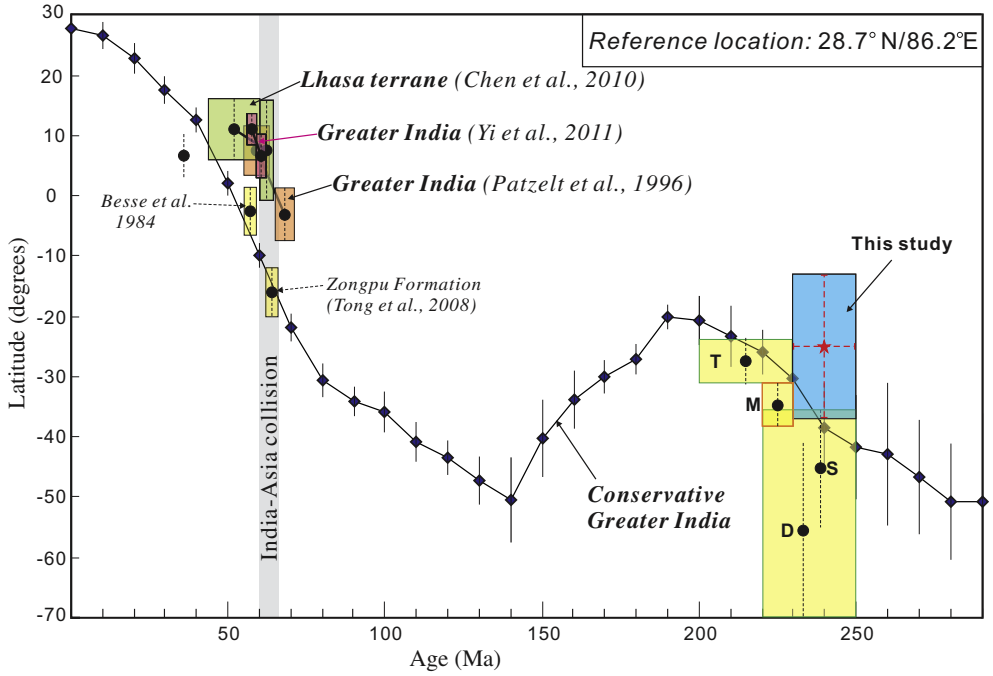


Fig. 7. Comparison of reported paleolatitudes (and 95% confidence limits) for reference location: 28.7°N/86.2°E (Besse et al. 1984; Chen et al., 2010; Patzelt et al., 1996; Tong et al., 2008; Yi et al., 2011; and this study) with the expected paleolatitudes according to the mean paleomagnetic poles of the Indian continent in Torsvik et al. (2008). Abbreviations: T – Thakkhola (Klootwijk and Bingham, 1980), M – Manang (Appel et al., 1991), S – Shiar (Schill et al., 2002), D – Wester Dolpo (Crouzet et al., 2003).

Table 4
Locality paleomagnetic poles from this study compared to expected directions from adjacent area of the Tethyan Himalaya (reference location 28.7°N/86.2°E).

Region	Age	Lat./Lon.	N(n)	Plat./Plon.	*A95*	Pλ	Test	Ref.
Western Dolpo	220–250	29.3/82.9	11	−1.5/284.5	17.1	55.4 S	FT	1
Thakkhola	200–230	28.7/83.7	3(233)	24.9/297.4	5.3	27.4 S	FT	2
Manang	220–230	28.6/84.0	2(40)	22.4/286.7	4.9	34.6 S	FT	3
Shiar	220–250	28.6/85.1	18	15.8/260.1	15.3	45.3 S	NO	4
Tingri	230–250	28.7/86.2	10	34.5/282.2	11.7	25.0 S	FT	This study
Gondwana	220 ± 10	28.7/86.2	35	28.7/297.5	2.0	25.3 S	5	
Gondwana	230 ± 10	28.7/86.2	28	25.3/294.2	2.6	29.7 S		5
Gondwana	240 ± 10	28.7/86.2	35	19.0/292.4	3.6	36.0 S		5
Gondwana	250 ± 10	28.7/86.2	38	12.8/292.9	4.4	41.1 S		5

Ref. – References: 1 – Crouzet et al. (2003); 2 – Klootwijk and Bingham (1980); 3 – Appel et al. (1991); 4 – Schill et al. (2002); 5 – Torsvik et al. (2008). Lat. – latitude, Lon. – longitude; Plat. – pole latitude; Plon. – pole longitude; N(n) – number of sites (samples) accepted; *A95* – the radius that the mean pole lies within 95% confidence; Pλ – paleolatitude for a reference point at 28.7°N/86.2°E on the Tethyan Himalaya FT – fold test. Mean paleomagnetic poles for Gondwana from 220 to 250 Ma.

since the Paleozoic without separation (e.g., Ali and Aitchison, 2005; Yi et al., 2011). This interpretation is also favored by the latest geologic observations. The 132 Ma old Comei–Bunbury large igneous province caused by the Kerguelen mantle plume in the process of breakup of eastern Gondwana (Zhu et al., 2008, 2009), is the parental magmas provenance of Lower Cretaceous Wölong volcaniclastics in the Tingri area (Hu et al., 2010) and Cr-spinels of the Tianba flysch in the Nieru Valley (Zhu et al., 2005b) in the Tethyan Himalaya Based on these observations, we propose that the Greater India has a large dimension (>1300 km) beyond its current northern margin prior to the Eurasia–India collision (e.g., Ali and Aitchison, 2005). This 1300 km dimension is also consistent with the total crustal shortening (>900 km) based on the balanced regional cross sections measurement in the Himalayan thrust belt (e.g., Robinson et al., 2006), rather than the Greater India margin underwent considerable extension in the Late Triassic (e.g., Draganits et al., 2005; Torsvik et al., 2009).

7. Conclusion

The Tulong Group marine rocks reveal well-defined remanence components that are carried by magnetite. The fold test strongly suggests a primary magnetization that was acquired during the Early to Middle Triassic. The Tulong Group pole overlaps with the India poles from 230 to 250 Ma which indicates the close affinity between the Tethyan Himalaya and cratonic India since the early Mesozoic. The 25.0° ± 11.7°S paleolatitude inferred by the Triassic Tulong Group is much further north than the expected for the Indian Shield, which indicates the original dimension of the Tethyan Himalaya prior to the India–Eurasia collision should exceed 1300 km. Our new results thus suggest crustal shortening between the Tethyan Himalaya and cratonic India has occurred on the order of 1300 km since the initial Eurasia–India collision at 65–60 Ma.

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